A Reality-Centric Framework for Safe and Adaptive Risk Forecasting

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Abstract

We develop a method that allows for the safe deployment of models derived from domain expertise even when misspecified and apply it to financial risk forecasting. This framework builds on the idea that domain-knowledge allows us to build models that can capture valuable dynamics of the problem, but as mere approximations of reality, they are inevitably misspecified which can lead to a breakdown of statistical guarantees, unaccounted forecasting errors, and a failure to adapt to changing circumstances. This paper provides a general wrapper that can be combined with any (black-box) model to provide forecasts that are safe, reliable, and accountable. Our methods achieve this by using the domain specific model to capture known dynamics and modelling residual dynamics as a learning problem which acts as a safety layer that learns to correct the errors in real time. We provide rigorous guarantees, proving that our method provides unconditionally valid Value-at-Risk and Expected Shortfall in the long run, irrespective of the data generating process. Through experiments on simulated and real-world data, we demonstrate that our method significantly outperforms existing methods and particularly excels in volatile environments.

1 Introduction

For many forecasting problems, we possess a rich body of knowledge how to solve them and capture them models. These models are powerful because they effectively characterise the essential dynamics of complex systems. Yet their practical value is hindered by a crucial limitation: as approximations of reality, they are inevitably misspecified. This can lead to a breakdown of statistical guarantees, unaccountable forecasting errors, and a failure to adapt to real world complexities. This becomes particularly problematic in high stakes forecasting tasks. Instead the ideal we strive for is a framework for reality-centric forecasting where models are not only effective but also probably reliable and accountable in all environments. This paper proposes such a framework, demonstrating how principles from online learning theory and game theoretic probability can be used to safely deploy any, possibly black box, base model.

This challenge unaccountable models is particularly critical in financial risk management. In the aftermath of the 2008 financial crisis the Basel accords mandate the forecasting of Value-at-Risk (VaR) and Expected Shortfall (ES) to determine the capital requirements of financial institutions. Where Value-at-Risk is the loss amount that you can expect to exceed only with some small predefined probability, and Expected Shortfall is the expected loss in the event such a loss occurs.

A significant step towards more robust and accountable risk forecasting comes from the work of Patton, Ziegel, and Chen (2019). Building on the theory of joint elicitability for Value-at-Risk (VaR) and Expected Shortfall (ES) developed by Fissler and Ziegel (2016), a proper loss function (known as the FZ loss) was introduced that is strictly consistent for the (VaR, ES) pair. This advancement is crucial because a strictly proper loss function ensures that minimising the expected loss incentivises forecasters to report the true (VaR, ES), thereby enabling objective forecast comparison and consistent estimation. Patton et al. (2019)) leveraged this idea to develop dynamic semiparametric models that directly minimize the FZ loss function to produce optimal forecasts for VaR and ES. The result is a dynamic semiparametric model that does not require parametric assumptions for the distributional assumptions of asset returns, representing a significant step towards more reality-centric methods in financial risk management. However, this framework still critically relies on parametric assumptions on the dynamics of these risk measures for consistent estimation. However as we noticed this is an unrealistic assumption. Therefore, this motivates our central question: can we provide a reality-centric method that guarantees valid tail risk forecasts when *both* the distributional and dynamic assumptions of the base model are misspecified?

In this paper, we propose a framework that adheres to these reality centric ideals using a two stage framework we call Model-Assisted Online Learning. First, we use a base model to normalise the returns data into a more stable sequence,on a bounded space. This stage encapsulates our existing domain knowledge. Second, we treat the resulting bounded sequence of residual dynamics as an arbitrary data stream and recast it as an online learning problem. We deploy a parameter-free, game theoretic online learning algorithm Orabona and Tommasi (2017) which acts as a safety layer. This layer learns to correct the prior's systematic errors in real time, while its learned parameters also serve as real-time diagnostics on the health of the underlying prior model. We provide rigorous guarantees for this framework, providing that our model provides unconditionally valid Value-at-Risk and Expected Shortfall forecasts in the long run, irrespective of the data generating process or the base model's initial misspecification. Furthermore, the choice of the COCOB algorithm inherently achieves the best of both worlds, it achieves convergence to the optimal conditional forecasts under stationary conditions while remaining fully adaptive to distribution shifts.

To evaluate our method, we employ alongside standard techniques, lesser known alternatives that respect our reality centric ideals. As many approaches rely on asymptotic theory and lesser known assumptions, that are sometimes swept under the rug. Therefore next to standard evaluation methods we use more reality centric evaluation methods to evaluate the forecasts and compare them. Specifically we employ universal inference (Wasserman, Ramdas, & Balakrishnan, 2020) which has strong connections to game-theoretic probability (Shafer & Vovk, 2019) the split likelihood ratio testing allows for finite sample testing Likelihood Ratio Tests under no regularity constraints. Through extensive experiments on real-world data, we demonstrate that this model assisted online learning framework outperforms existing approaches and particularly adapts fast to structural breaks.

This paper is built on the principles of what we term Reality-Centric modelling. The central idea is that predictive systems should not only be effective, but also demonstrably reliable and accountable, such that they can effectively be deployed in the complexities of the real world. This is especially relevant with the rising popularity of 'black-box' models, whose effectiveness in certain tasks is often overshadowed in critical tasks Rudin (2019). Fields like Explainable AI have similar goals Ribeiro, Singh, and Guestrin (2016) however they try to achieve this by making these models more interpretable. The reality centric focus shares this ideal, but also approaches this from a different angle, namely while we might not be able to interpret them we hope to make them safe in such a way that we can still trust them. The specific framework proposed in this paper then aims to live up to these ideals with the additional reality centric ideal that it respects the known knowledge and methods that have already been developed and allows to use them in an unrestricted fashion by providing a general wrapper that strives to make any base model safe.

Another method that has been tried for Value at Risk forecasting Fantazzini (2024) is Adaptive Conformal Inference (Gibbs & Candès, 2024). Conformal prediction (Angelopoulos & Bates, 2022; Vovk, 2012) is a distribution free framework

2 Tail Risk Forecasting

Consider the sequential problem of forecasting the risk of financial returns. Let $\{Y_t\}_{t=1}^{T}$ be a sequence of asset returns, where positive positive values denote profits and negative values denote losses. We assume that at each time t, all information available up to the previous period is contained in the information set $\mathscr{F}_{t-1} = \sigma(Y_1, \ldots, Y_{t-1})$ which is a natural filtration. Let the conditional distribution of Y_t given $\mathscr{F}_{t-1}$ be denoted by $F_t(\cdot)$ which we assume to be strictly increasing and has finite mean. Our objective is to produce one-step-ahead for the two key risk measures: Value at Risk (VaR) and

Expected Shortfall (ES), for a given tail probability level $\alpha \in (0, 0.5)$, which is typically small (e.g. $0.01, 0.025, 0.05$).

The Value at Risk represents the maximum potential loss that is expected to be exceeded with a probability α for each day. Formally

Definition 1 (Value-at-Risk) *The α-level VaR at time t, for Y_t, denoted $v_{y,t}$ is the α-quantile of the conditional distribution F_t*

$$v_{y,t} := F_t^{-1}(\alpha)$$

Definition 2 (Expected Shortfall) *The α-level ES at time t, for Y_t denoted $e_{y,t}$ is the expected value of the return conditional on it being less than equal to the VaR at time t. Formally,*

$$e_{y,t} := \mathbb{E}\left[Y_t \mid Y_t \leq v_{y,t}, \mathscr{F}_{t-1}\right]$$

The ES measures the average severity of losses when bellow the VaR, thereby providing a more detailed view of tail risk. By definition, ES is always more extreme than VaR, so $e_{y,t} \leq v_{y,t}$.

2.1 Elicitability and The Fissler-Ziegel Loss

An important concept in assessing and evaluating forecasts is *elicitability*. A functional of a distribution (like mean, median, quantile) is elicitable if it is the unique minimiser of the expectation of some loss function. For example Savage (1971) shows that the mean is elicitable using the squared error.

Only for an elicitable functional does a consistent scoring function exist ensuring that a forecasters unique optimal strategy is to report the true prediction for the forecast. Without it, any chosen evaluation metric could favour a biased forecast making comparisons uninformative.

For our tail risk forecasting problem however the functional VaR (the quantile of a distribution) is elicitable on its own through the hack loss (also known as the tick or pinball loss). It is established that expected shortfall is not (Gneiting, 2011).

A breakthrough by Fissler and Ziegel (2016) resolved this issue by proving that while ES is not elicitable on its own, the vector (VaR, ES) is *jointly elicitable*. For this specific purpose they provide a family of loss functions that are strictly consistent for the pair $(v_{y,t}, e_{y,t})$, of which any member can be used to consistently estimate and evaluate the VaR and ES without needing to specify the conditional distribution F_t.

Specifically the Fissler-Ziegel (FZ) family of loss functions $L(Y, v, e)$ is specified for two functions G_1 and G_2:

$$L_{FZ} = (\mathbb{1}\{Y \leq v\} - \alpha)\left(G_1(v) - G_1(Y) + \frac{1}{\alpha}G_2(e)v\right) - G_2(e)\left(\frac{1}{\alpha}\mathbb{1}\{Y \leq v\}Y - e\right) - \mathcal{G}_2(e)$$

where G_1 is non-decreasing, G_2 is strictly positive and strictly increasing, and $\mathcal{G}_2$ is a function such that it's derivative $\mathcal{G}_2' = G_2$.

This theoretical advancement was then applied to dynamic tail risk modelling by Patton et al. (2019). Who use the FZ loss function in Generalised Autoregressive Score (GAS) models, and GARCH model to directly estimate the parameters this is motivated that if the dynamics are correctly specified (Quasi)-Maximum Likelihood Estimation is almost certainly best. However if the model is misspecified then estimating the parameters using the FZ loss yields the parameters that deliver the best VaR and ES forecasts. What is left is a semiparametric model that assumes parametric assumptions for the dynamics but makes n assumptions about the distribution of returns

This approach however still requires that the dynamics are correctly specified for any consistent estimation and further provide no validity guarantees. For instance, their modle might assume a GARCH like structure for the evolution of the risk measures however if this dynamic structure is wrong the consistency guarantees are lost and while still the best

[INSERT PART ABOUT consistent estimation]

3 An adaptive safety layer for parametric models

The framework we propose is a two stage process to designed to safely. We first use a structural prior to simplify the problem, and then deploy an adaptive safety layer to correct the prior's residual errors. This seperation allows the framework to be both effective, by using established domain knowledge, and safe, by

3.1 Stage 1: Normalisation using Parametric model

The framework begins by assuming the existence of a *prior model* $\mathscr{M}_t$. This prior, at each time t, provides an order-preserving transformation $\hat{F}_t : \mathcal{Y} \to \mathcal{U}$ for the return Y_t mapping it from its original space to the bounded domain $\mathcal{U} = [0,1]$[1]. A natural candidate for such a transformation is the conditional Cumulative Distribution Function (CDF), we will use this choice for the remainder of this paper. Note that the prior can be any model such as a GARCH class model, but this notion can also extend to models which are not conventionally associated with parametric distributional assumptions by parsing the residuals to through a well chosen CDF anyways, however arbitrary this may be. e.g. normal(0,1).

This first stage serves three purposes.

First, its primary role is to homogenise the observations by filtering out primary dynamics. By applying a domain-specific model like GARCH, we essentially normalise the raw financial returns. This process them simplifies the complex unbounded process into a more stationary residual sequence in the bounded domain. The aim is to make it easier for subsequent online learning, irrespective of what residual dynamics remain.

Second, this transformation ensures that the problem becomes bounded. Specifically, it guarantees that the distance between any forecasted variable $\boldsymbol{x}_t = (v_{y,t}, e_{y,t})$ and their true underlying latent targets $\boldsymbol{x}_t^\star = (v_{y,t}^\star, e_{y,t}^\star)$ remain within a quantifiable, finite range. We will show that this is a useful property for many online learning algorithms, and allows us to derive the long-run performance guarantees.

Third, this stage provides a way to reason about the correctness of our prior and provides a real-time diagnostic on the health.

- In the ideal, best case, where the prior model perfectly captures both the distributional characteristics and the dynamics of returns, then the residual sequence would be independent and identically distributed (iid) and uniform on $[0, 1]$.
- In the second best case, where the dynamics are correctly specified but only its distributional assumptions are flawed (e.g. assuming gaussian, but errors are truly fat-tailed) then the residual sequence $\{U_t\}$ would exhibit a stationary but non-uniform distributions. Importantly, in these best two cases the true VaR and ES function are stable over time.
- More realistically, both the distributional assumptions as the dynamics are incorrectly specified, leading the residual sequence $\{U_t\}$ to become non stationary, and the true underlying $(v_{u,t}^\star, e_{u,t}^\star)$ will drift over time.

This last point is particularly important as observing the behaviour of VaR and ES in the original return space is not inherently informative, as these measures are expected to change dynamically in financial markets. However, observing any drift in the functional $\mathrm{s}(v_{u,t}, e_{u,t})$ within the bounded space is an immediate signal of model misspecification. This provides a powerful economic analysis tool for real-time diagnostics of the prior model, revealing precisely where and how our theoretically grounded model might be failing.

Ultimately this first stage aims to yield a more well behaved residual sequence which the adaptive safety layer then learns to correct for dynamic misspecification in real time, to transform the imperfect but useful prior into a safe and robust forecasting system.

[1] This can be any convex bounded domain, however without loss of generality we assume this to be [0,1].

3.2 Stage 2: Online Learning as a safety layer

After the transformation from the first stage, the tasks becomes to make predictions for the bounded residual sequence $\{U_t\}$. Crucially, after incorporating all our knowledge in the first stage (through the prior model $\mathscr{M}_t$), we impose no further assumptions on this transformed sequence. We treat it as an arbitrary, potentially non-stochastic, or even weakly adversarial data stream. For this reason, we switch the notation from U_t to u_t to emphasise its potentially deterministic nature. This setup is highly familiar in the field of Online Learning, which provides powerful tools for making sequential predictions under such minimal assumptions.

The standard setup in online learning is as follows:

Algorithm 1 Standard Online Learning Setting

for *t = 1* **to** *T* **do**

1. the learner chooses a forecast $\boldsymbol{x}_t = (v_{u,t}, e_{u,t}) \in \mathcal{K} \subseteq \mathcal{U}$
2. An adversary chooses loss function $\ell_t : \mathcal{K} \to \mathbb{R}$, the learner pays $\ell_t(\boldsymbol{x}_t)$
3. the learner updates its model to choose $\boldsymbol{x}_{t+1}$

Where the loss $\ell_t(\boldsymbol{x}_t) := \ell_t(u_t; \boldsymbol{x}_t)$ means that when the adversary chooses loss ℓ_t he reveals outcome u_t. Specific to our setting, the online learning problem is to learn the parameters $(v_{u,t}, e_{u,t})$ that track the α-level VaR and ES of the sequence $\{Y_t\}$ in the bounded domain. A Natural way to do this is using Projected Stochastic Gradient Descent (SGD), as outlined bellow

Algorithm 2 Projected Subgradient Descent

Require: Convex set $\mathcal{K} \subseteq \mathcal{U}$, $\boldsymbol{x}_1 \in \mathcal{K}$, $\eta_1, \ldots, \eta_T > 0$

for *t = 1* **to** *T* **do**

1. output $\boldsymbol{x}_t$
2. Receive $\ell_t : \mathcal{K} \to \mathbb{R}$ and pay $\ell_t(\boldsymbol{x}_t)$
3. Set $\boldsymbol{g}_t = \partial \ell_t(\boldsymbol{x}_t)$
4. $\boldsymbol{x}_{t+1} = \Pi_{\mathcal{K}}(\boldsymbol{x}_t - \boldsymbol{\eta}_t \boldsymbol{g}_t) = \arg\min_{\boldsymbol{y} \in \mathcal{K}} \|\boldsymbol{x}_t - \boldsymbol{\eta}_t \boldsymbol{g}_t - \boldsymbol{y}\|$

where $\boldsymbol{g}_t \in \partial \ell_t(\boldsymbol{x}_t)$ is a subgradient of the loss, $\boldsymbol{\eta_t}$ is the step size, and $\Pi_{\mathcal{K}}$ is the projection onto the feasible space $\mathcal{K}$.

3.2.1 Operating in the bounded domain and the choice of Loss function

A critical design choice in our framework is to model the risk measures directly in the bounded domain by minimising the FZ loss function on the residual sequence $\{u_t\}$ rather than on the original returns $\{Y_t\}$. [Section with r] This generally leads to an underestimation of the true Expected Shortfall due to the convexity of the CDF transformation. However this is

For our online learning problem, we purposely choose a specific form of the Fissler-Ziegel loss function to yield stable and simple gradients. For the VaR Component we choose $G_1(v) = v$, which yields $G_1'(v) = 1$. And for the ES component we choose $G_2(e) = e$, which is strictly increasing and strictly positive on the domain. This implies its integral $\mathcal{G}_2(e) = \frac{1}{2}e^2$.

Substituting these choices into the general FZ formula, where Y_t is now our bounded outcome u_t, we obtain the specific FZ loss function $L_{stable}(u_t; v_{u,t}, e_{u,t})$:

$$L(u_t; v_{u,t}, e_{u,t}) = \Big(\mathbb{1}\{u_t \leq v_{u,t}\} - \alpha)(v_{u,t} - u_{u,t} + \frac{e_{u,t} v_{u,t}}{\alpha}\Big) - e_{u,t}\left(\frac{1}{\alpha}\mathbb{1}\{u_t \leq v_{u,t}\} - e_{u,t}\right) - \frac{1}{2}e_{u,t}^2$$

From this specific loss function $L_{stable}(u_t, v_{u,t}, e_{u,t})$, some simple simple derivations yield the subgradients $\boldsymbol{g}_t = (\partial L/\partial v_{u,t}, \partial L/\partial e_{u,t})$ for our Projected Stochastic Gradient Descent Algorithm:

$$g_{v,t} = \frac{\partial L}{\partial v_{u,t}} = (\mathbb{1}\{u_t \leq v_{u,t}\} - \alpha)\left(1 + \frac{e_{u,t}}{\alpha}\right)$$

Notice that this gradient is the tick loss scaled by a factor of $(1 + \frac{u_{u,t}}{\alpha})$.

$$g_{e,t} = \frac{\partial L}{\partial e_{u,t}} = e_{u,t} - v_{u,t} + \frac{v_{u,t} - u_t}{\alpha} \mathbb{1}\{u_t \le v_{u,t}\}$$

While these are the exact gradients of the L_{stable} loss, for the purpose of the Projected Subgradient Descent updates, we employ a normalised version of the VaR gradient by dropping the $(1 + \frac{e_{u,t}}{\alpha})$ term. This is common practise, when the loss L_{stable} is quasi-convex, and as the scaling factor is consistently positive this does not alter the sign of the gradient, and hence simplifies the update rule without changing its root. This leaves us with the following normalised projected subgradient descent update rule:

$$v_{u,t+1} = \Pi_{\mathcal{K}} \left(v_{u,t} - \eta_{v,t} \left(\mathbb{1}\{u_t \le v_{u,t} - \alpha\} \right) \right)$$
$$e_{u,t+1} = \Pi_{\mathcal{K}} \left(v_{u,t} - \eta_{e,t} \left(e_{u,t} - v_{u,t} - \frac{u_t - v_{u,t}}{\alpha} \mathbb{1}\{u_t \le v_{u,t}\} \right) \right)$$

Notice that the normalised subgradients that we are left with are the identification functions for the VaR, and the ES in the bounded domain. This precise formulation, using gradients derived directly in the bounded u_t space offers powerful interpretations of our algorithm.

First, the Normalised Subgradient Descent algorithm, employing these specific subgradients is mathematically equivalent to the Robbins-Monro Stochastic Approximation algorithm (Robbins & Monro, 1951). This connection is significant as it provides a well understood mechanism for iterative convergence towards the optimal parameters that minimise the expected FZ loss, and allows to provide following theoretical guarantees that support the reality centric framework.

Proposition 1 (VaR Unconditional Coverage) *For any sequence of observations $\{u_t\}_{t=1}^T$ and for any constant step size $\eta_v > 0$, the sequence of VaR estimates $\{v_{y,t}\}$ constrained to the bounded domain with diameter D. produced by the SGD VaR update rule satisfies*

$$\left| \frac{1}{T} \sum_{t=1}^{T} \left(\mathbb{1}\{u_t \le v_{u,t}\} - \alpha \right) \right| \le \frac{D}{\eta_v T}$$

Therefore, in particular as F_t is a strictly increasing function

$$\lim_{T \to \infty} \frac{1}{T} \sum_{t=1}^{T} \mathbb{1}\{Y_t \le v_{y,t}\} \overset{a.s}{=} \alpha$$

In words, this guarantee assures that even under an arbitrary data stream the proportion of VaR violations approaches the nominal target rate α in the long run

Proposition 2 (ES Weak Unconditional Validity) *For any sequence of observations $\{u_t\}_{t=1}^T$ and for any constant step size $\eta_e > 0$, the sequence of ES estimates produced by the US update rule satisfies*

$$\left| \frac{1}{T} \sum_{t=1}^{T} \left[e_{u,t} - v_{u,t} + \frac{v_{u,t} - u_t}{\alpha} \mathbb{1}\{u_t \le q_{u,t}\} \right] \right| \le \frac{D}{\eta_e T}$$

Therefore, as

$$\lim_{T \to \infty} \frac{1}{T} \sum_{t=1}^{T} e_{u,t} \overset{a.s}{=} \lim_{t \to \infty} \frac{1}{T\alpha} \sum_{t=1}^{T} u_t \mathbb{1}\{u_t \le v_{u,t}\}$$

In words, the average forecasted $e_{u,t}$ approaches the sample estimator of the ES (conditional on being less than the forecasted VaR rather than the true VaR) in hindsight.

Proofs for both propositions are deferred to Appendix A.

Beyond these formal guarantees, another powerful interpretation is that the SGD update rule can be seen as dynamically tracking the true, or nominal, α-level VaR and ES targets by attempting to find alternative α levels for VAR($\alpha_{v,t}$), and ES($\alpha_{e,t}$) that satisfy the properties of the nominal α-VaR and ES. (Setting $\alpha_{v,t} := v_{u,t}$, and $\alpha_{e,t} := e_{u,t}$). This adaptive tracking is a core element of the safety layer, allowing the model to self-correct in real time.

Remarks on the approximation and its implications

It is important to acknowledge that directly estimating the VaR and ES on the bounded residuals is just an approximation of the true α-level VaR and ES of the unbounded returns Y_t. This simplification greatly enhances the tractability and practical feasibility of the online learning framework as it avoids the need to compute the conditional density as would be required by the chain rule. This does introduce some nuances. For instance this approximation generally leads to a subtle underestimation of the expected shortfall due to the general convexity in the left tail of the CDF (Jensen's Inequality). However our empirical results demonstrate that despite this approximation, our models often pass the specification test for ES. This suggests that the appriximation is robust and sufficient for reliable forecasting. Moreover, this approximation tends to improve when forecasting into the tails of the distribution where the local convexity of the CDF typically decreases. We have computed derivations for common distributions to quantify this bias in Appendix D. A more detailed discussion on the implications and the theoretical nuances of this approximation, particularly concerning the stability of ES, is provided in Section 7 (Discussion).

3.3 Choosing the learning rate schedule

The choice of the learning rate schedule for $(\eta_{v,t}, \eta_{e,t})$ provides a tradeoff between stability and adaptiveness. A large learning rate allows for the quick adaption to distribution shifts and the results of propositions 1 and 2 seem to suggest rapid convergence. However, in practise this can lead to oscillation and considerable volatility in the estimates $(v_{u,t}, e_{u,t})$ making them serially dependent.

Furthermore we need a justification for the choice of our updates. Notably consider the Newton-Raphson update

$$x_t = x_{t-1} - \boldsymbol{\eta} \left[H^\star\right]^{-1} \boldsymbol{g}_{t-1}.$$

Using the Hessian of the expected loss, evaluated at the true solution $(v_t^\star, e_t^\star)$ is diagonal

$$H^\star = \partial^2 \mathbb{E}(L_{\text{stable}}) = \begin{bmatrix} \frac{f_t(v_t)}{\alpha} & 0 \\ 0 & 1 \end{bmatrix}$$

In the idealised uniform case: $f_t(v_t) = f_t(\alpha) = 1$. Hence the inverse hessian trivially becomes

$$\left[H^\star\right]^{-1} = \begin{bmatrix} \alpha & 0 \\ 0 & 1 \end{bmatrix}$$

This provides an intuitive results: Namely, that the Newton-Raphson update is still a stochastic gradient descent algorithm but the learning rate of the VaR update is automatically scaled by the target level α. For smaller α, the updates are even smaller and make the process more stable in the tails. Simultaneously it provides an intuitive result that if we want to choose one global learning rate parameter for parsimony, then the effective learning rate η_e should be smaller with a factor α. For the purpose of the experiments in this paper, a global learning rate of 0.02 is chosen. This was chosen because it was able to provide relatively stable trajectories for $v_{u,t}$ and still adapts to distribution shifts.

3.4 Parameter Free Adaptive Online Learning

The performance of the Stochastic Gradient Descent algorithm described above critically depends on the choice of the step size sequence, $\{\eta_t\}$. Manually tuning this hyperparameter is a significant practical bottleneck and goes against the philosophy of creating a truly adaptive and autonomous system. A core goal of reality-centric modeling is to reduce such external dependencies, leading to algorithms that are robust and require minimal intervention. Therefore, we seek a parameter-free update rule that can implicitly adapt its learning rate to the structure of the problem. Furthermore, we desire an algorithm that achieves the best of both worlds: If the residual sequence happens to be stationary it should converge to the true conditionally valid risk measures, yet remain adaptive and capable of tracking drift if te sequence is non-stationary.

To achieve this, we turn to the state-of-the art in parameter-free optimisation and employ employ the COntinuous COin Betting (COCOB) algorithm, proposed by Orabona and Tommasi (2017). This algorithm reframes the online learning problem as a sequential betting game. At each step, the algorithm "bets" a fraction of its capital on the sign of the next gradient. Then using results for optimal bet sizing from Information Theory (Kelly, 1956) This game-theoretic perspective allows for the derivation of an update rule that is not only parameter-free but also optimal under very general conditions. The pseudocode for the COCOB algorithm adapted for our two coordinate problem is presented in algorithm 3.

Algorithm 3 The COCOB Algorithm as an Adaptive Safety Layer

Require: Initial parameters $\boldsymbol{x}_1 = (v_{u,1}, e_{u,1}) \in \mathcal{K}$. Nominal quantile level $\alpha \in (0, 0.5)$.
▷ *Initialize algorithm state variables*
1: Set gradient bounds: $L_v \leftarrow 1$, $L_e \leftarrow 0.5/\alpha$.
2: Initialize for VaR coordinate: $G_{0,v} \leftarrow L_v$, $Reward_{0,v} \leftarrow 0$, $\theta_{sum,v} \leftarrow 0$.
3: Initialize for ES coordinate: $G_{0,e} \leftarrow L_e$, $Reward_{0,e} \leftarrow 0$, $\theta_{sum,e} \leftarrow 0$.
4: Store initial parameters: $v_{u,start} \leftarrow v_{u,1}$, $e_{u,start} \leftarrow e_{u,1}$.
for $t = 1$ **to** T **do**
5:
Forecast with current parameters $\boldsymbol{\theta}_t = (v_{u,t}, e_{u,t})$.
6: Receive observation $u_t \in [0, 1]$.
▷ *VaR Update Step*
7: Calculate VaR gradient: $g_{v,t} \leftarrow \mathbb{1}\{u_t \leq v_{u,t}\} - \alpha$.
8: Update VaR statistics:
9: $G_{t,v} \leftarrow G_{t-1,v} + |g_{v,t}|$.
10: $Reward_{t,v} \leftarrow \max(Reward_{t-1,v} + g_{v,t} \cdot (v_{u,t} - v_{u,start}), 0)$.
11: $\theta_{sum,v} \leftarrow \theta_{sum,v} - g_{v,t}$.
12: Calculate next VaR parameter: $v'_{u,t+1} \leftarrow v_{u,start} + \frac{\theta_{sum,v}}{G_{t,v}}(L_v + Reward_{t,v})$.
▷ *ES Update Step*
13: Calculate ES gradient: $g_{e,t} \leftarrow e_{u,t} - v_{u,t} + \frac{v_{u,t} - u_t}{\alpha}\mathbb{1}\{u_t \leq v_{u,t}\}$.
14: Update ES statistics:
15: $G_{t,e} \leftarrow G_{t-1,e} + |g_{e,t}|$.
16: $Reward_{t,e} \leftarrow \max(Reward_{t-1,e} + g_{e,t} \cdot (e_{u,t} - e_{u,start}), 0)$.
17: $\theta_{sum,e} \leftarrow \theta_{sum,e} - g_{e,t}$.
18: Calculate next ES parameter: $e'_{u,t+1} \leftarrow e_{u,start} + \frac{\theta_{sum,e}}{G_{t,e}}(L_e + Reward_{t,e})$.
▷ *Project onto the parameter space*
19: $\boldsymbol{\theta}_{t+1} \leftarrow \Pi_{\mathcal{K}}\left((v'_{u,t+1}, e'_{u,t+1})\right)$, where $\mathcal{K} = \{(v, e) \mid 0 < e \leq v < 0.5\}$.
20:

The algorithm maintains a capital K_i for each parameter. Where if the wealth is large it takes large steps and vice versa. The update rule for the the parameters (Line 5) is a function of the cumulative sum of past gradients scaled by the current wealth. Intuitively, if the gradients consistently point in the same direction, that is is the gradients are of the same sign because the risk measures are for example underestimating the true risk, then the sum of gradients θ_i grows in magnitude, and the wealth K also grows causing the algorithm to take larger, more aggressive steps. Conversely, if the gradients flip signs, this indicates that the forecasts have surpassed the optimum and, and are oscillating around the target, with as a result that the wealth decreases, causing the algorithm to slow down and take smaller steps. This allows COCOB to speed up when it is far from the optimum and slow down as it is close, which allows COCOB to provide optimal convergence properties. the FZ loss is quasi convex and locally strongly convex near its optimum the COCOB algorithm should achieve optimal convergence guarantees if the initial parameters are well specified. Given the nature of our base model we will take heuristic $v_{u,0} = \alpha$ and $e_{u,0} = \alpha/2$

We must provide the gradient bounds L_v, and L_e. We derive these by analysing the gradients over the bounded domain. For the VaR gradient is $g_{v,t} = 1u_t \leq v_{u,t} - \alpha$. its absolute value is bounded by $\max\{|1-\alpha|, |-\alpha|\}$. Given that $\alpha \in (0, 0.5)$ we set $L_v = 1 - \alpha$. The ES gradient is $e_{u,t} - v_{uu,t} + \frac{v_{u,t} - u_t}{\alpha}\mathbb{1}\{u_t \leq v_{u,t}\}$. By analysing the extrema we find that its value is bounded within $[-0.5, 0.5/\alpha]$. Hence we set the bound $L_e = 0.5/\alpha$.

4 Evaluation

Having proposed a framework for safe and adaptive risk forecasting, we now turn to its empirical evaluation. Our reality-centric philosophy is that not only our forecasting models should be safe to use but also our evaluation methods should follow these ideals for correct decision making. Many standard statistical tests for forecast evaluation rely on asymptotic theory and modelling assumptions that may not hold in the non- stationary world of the financial markets. In this section, we therefore detail several evaluation method that are lesser known but provide a more reliable assessment of model performance in finite samples. We will deploy these next to the standard tests for VaR and ES assesment.

4.1 Finite Sample Goodness of fit testing

Standard approaches for evaluating the the goodness of fit of VaR and ES forecasts include the unconditional coverage test of Kupiec (1995), and independence test of Christoffersen (1998) , and the DQ and DES MZ regression tests of Engle and Manganelli (2004). However. all of these tests rely on asymptotic theory. For example Kupiec and Christopherson are both Likelihood Ratio Tests (LRT).

The general theory for Likelihood Ratio tests is as follows. To test the null hypothesis $H_0 : \theta^* \in \Theta_0$ vs the alternative $H_1 : \theta^* \in \Theta_1 \supset \Theta_0$. Then under regularity conditions for the models by Wilk's Theorem:

$$2 \log \frac{\mathscr{L}(\hat{\theta}_1)}{\mathscr{L}(\hat{\theta}_0)} \to \chi^2_d$$

under H_0. where $\mathscr{L}(\theta) := \Pi_{i=1}^n p_\theta(Y_i)$ is the likelihood function. $\hat{\theta}_{0/1}$ is the MLE under $\Theta_{0/1}$. d is the difference in dimensionality between Θ_0, Θ_1. Then the LRT rejects if

$$2 \log \frac{\mathscr{L}(\hat{\theta}_1)}{\mathscr{L}(\hat{\theta}_0)} \geq \chi^2_{,d}(1-\alpha).$$

Then under regularity conditions $\mathbb{P}_{H_0}(\text{rejection}) \leq \alpha + o_p(1)$.

The reason why the likelihood ratio test only yields asymptotic guarantees is that the MLE for the alternative is estimated and evaluated on the *same* dataset. This leads to an inflated likelihood ratio. To overcome this limitation and achieve finite sample control, we turn to Universal inference (Wasserman et al., 2020), which proposes the Split Likelihood Ratio test. This methodology avoids the 'overfitting' issue by estimating the MLE of the alternative on one part of the dataset and evaluating it on another seperate dataset. The Split Likelihood Ratio Test proceeds as follows. First split the data in two parts $\mathcal{D}_0, D_1$. then the split LRT rejects if

$$2\text{log}\frac{\mathscr{L}_0(\hat{\theta}_1)}{\mathscr{L}_0(\hat{\theta}_0)} \geq 2\log(1/\alpha)$$

Where $\mathscr{L}_0(\theta) := \Pi_{i \in \mathcal{D},} p_\theta(Y_i)$ is the likelihood on $\mathcal{D}_0$. $\hat{\theta}_1$ is then an estimator under Θ_1 on $\mathcal{D}_1$, and $\hat{\theta}_0$ is the MLE under Θ_0 on $\mathcal{D}_0$. As shown in Wasserman et al. (2020) , this test guarantees that under any sample size and no regularity condition that the probability of a type I error is bounded by alpha: $\mathbb{P}_{H_0}(\text{reject}) \leq \alpha$.

Proof: Under H_0, such that $\arg\max_\theta \mathscr{L}_0(\theta) = \theta^* \in \Theta_0$ it holds that

$$\mathbb{E}_{\theta^* \in H_0}\left[\frac{\mathscr{L}_0(\hat{\theta}_1)}{\mathscr{L}_0(\hat{\theta}_0)}\right] \leq \mathbb{E}_{\theta^* \in H_0}\left[\frac{\mathscr{L}_0(\hat{\theta}_1)}{\mathscr{L}_0(\theta^*)}\right] = \Pi_{i \in \mathcal{D}_0} \int \frac{p_{\hat{\theta}_1}(y)}{p_{\theta^*}(y)} p_{\theta^*}(y) dy = 1.$$

Therefore by Markov's inequality, $\mathbb{P}_{H_0}\left(\frac{\mathscr{L}_0(\hat{\theta}_1)}{\mathscr{L}_0(\hat{\theta}_0)} > 1/\alpha\right) \leq \alpha.$ □

This elegant and simple result provides a general and robust method for hypothesis testing. In our empirical evaluation we will report this, next to the asymptotic equivalent, to employ finite-sample version of the Kupiec Unconditional Coverage test and the Conditional coverage test.

4.2 Assumption Free Forecast Comparison

One we have generated forecasts from competing models, a logical next step is to determine which model performed better. Diebold and Mariano (1995) first popularised the testing for equal predictive accuracy over a set of predictions. However their test relies on asymptotic results and assumes the stationarity of the score differentials. These are assumptions that are often swept under the rug in practise, particularly in the nonstationary environment of financial markets. Instead, also for forecast comparison reality centric methods need reliable guarantees in all environments, especially in nonstationary ones.

To formalise the requirement, we seek an evaluation framework that is itself robust. the ideal comparison is based on a strictly proper scoring rule $S(\boldsymbol{x}, Y)$, which ensures. As previously discussed, a scoring rule is strictly proper if a forecasters expected score (the inverse of the loss) is uniquely maximised by their true belief. That is, $\arg\max_{\boldsymbol{x}^*} \mathbb{E}_{Y\sim D(\boldsymbol{x}^*)}[S(\boldsymbol{x}, Y)] = \boldsymbol{x}^*$, where $D(\boldsymbol{x}^*)$ is a conditional distribution with true risk measures $\boldsymbol{x}^* = (v^\star_{y,t}, e^\star_{y,t})$

Similary as the methods we have discussed before, more robust, assumption-free approach to forecast comparison stems from the field of game-theoretic probability (Shafer & Vovk, 2019) and the construction of confidence sequences (Howard, Ramdas, McAuliffe, & Sekhon, 2021). Choe and Ramdas (2024) propose a sequential forecast comparison framework for binary prediction that provides time-uniform guarantees with remarkably no assumptions on the data generation but that the score function (the negative of the evaluation loss) is bounded. We adapt their framework to the continuous setting for VaR and ES forecast comparison. They propose the following game protocol for sequential forecast comparison:

1. At each time t, information $\mathscr{F}_{t-1}$ is revealed.
2. Two competing forecasters provide their (VaR, ES) forecasts, $\boldsymbol{x}_{1,t}$ and $\boldsymbol{x}_{2,t}$
3. Reality chooses a conditional distributon r_t on $\mathcal{Y}$, which is strictly increasing and has finite mean.
4. The outcome $Y_t \sim r_t$ is revealed.

At each step, this gives the score differential of game t, $\hat{\delta} := S(\boldsymbol{x}_{1,t}, Y_t)$. Its conditional expectation $\delta_t := \mathbb{E}_{r_t}[\delta \mid \hat{\mathscr{F}}_{t-1}]$ and cumulative average differential $\Delta_t = \frac{1}{t}\sum_{i=1}^{t}\delta_t$. The cumulative score differential $S_t = \sum_{i=1}^{t}(\hat{\delta}_t - \delta_t)$ is then a martingale for any $\boldsymbol{x}_{1,t}, \boldsymbol{x}_{2,t}$, and r_t, as $\mathbb{E}_{r_t}[\hat{\delta}_t \mid \mathscr{F}_{t-1}] = \delta_t$. Choe and Ramdas (2024) show that if the loss differential is bounded by L because the scoring function is bounded over $\mathcal{Y}$, then using the work of Howard et al. (2021) this structure can be used to create a non-negative super martingale with, with the defining property $\mathbb{E}[S_t \mid S_{t-1}] \leq S_{t-1}$ The result is a process of the form

$$K_t(d) := \exp\left(\sum_{i=1}^{t}\lambda_t(\hat{\delta} - d) - \Psi_E(\lambda_t)\hat{V}_t\right),$$

where d is a hypothesised mean difference, such that $K_t(\Delta_t)$ is a non-negative supermartingale. The sequence is initialised at $K_0 = 1$. and $\lambda_t \in [-1, 1]$ is adaptively chosen based on past data for optimal growth.

Ville's inequallity (Ville, 1939), which is a time uniform equivalent of Markov's inequality for non-negative super martingale provides

$$\mathbb{P}_{r_t}(\exists t \in \mathbb{N} : K_t \geq 1/\alpha) \leq \alpha.$$

By inverting this test for all possible values of d, this yields a confidence sequence $C_t := \{d \in [-L, L] : K_t(d) < 1/\alpha\}$ which contains the true differential Δ_t with probability at least $1 - \alpha$, simultaneously over all time t.

The power of this approach is its generality. It places no assumptions on how the forecasts $\boldsymbol{x}_{1,t}$, and $\boldsymbol{x}_{2,t}$ are generated. Furthermore Choe and Ramdas (2024) show that the game protocol 4.2 can

For our experiments we choose the pragmatic solution of choosing the bound L based on the maximum absolute score difference on an initial in sample period used for fitting the base model described in Section REF, following a burn in period of 1000 observations to prevent noise in the initial scores to effect this bound. This bound is then used to truncate future scores. It must be acknowledged that this technically voids the non-asymptotic guarantees for the original FZ0 loss, as it changes the quantity being evaluated. However for its simplicity, it is deemed that this is a necessary

In the empirical evaluation in Section 6.2, we will report the final interval of the confidence sequence, C_T, alongside the confidence interval form the traditional Diebold-Mariano test to assess the effectiveness of this assumption-free, finite sample against the asymptotic counterpart.

5 Connection to Adaptive Conformal Prediction

A particularly effective method for producing sequential prediction sets (in particular one-sided intervals of the form $[v_t, \infty)$, needed for VaR forecasting) is adaptive conformal prediction(Gibbs & Candes, 2021). This is an adaptation of traditional Conformal prediction Angelopoulos and Bates (2022); Vovk, Gammerman, and Shafer (2005) which promises to produce valid prediction intervals without making any distributional assumptions. and is able to produce statistical guarantees of the form:

$$\mathbb{P}(Y_t \in C_t) \geq 1 - \alpha.$$

Note however that the only assumption this framework relies on is exchangeability of the data. In time-series however the marginal distribution of stock returns is not expected to be stationary. For this purpose adaptive conformal prediction adapts this procedure by tracking the sample quantile that models the distribution shift through:

$$q_{t+1} = q_t + \eta(\mathbb{1}\{Y_t \notin C_t(X_t)\} - \alpha)$$

This is precisely the SGD update for the α-quantile of the observations, and is identical to the VaR update rule in our own algorithm when operating in the bounded domain, where the non-conformity score is simply the identity function, $s(u) = u$.

Note however that the guarantee provided by (adaptive) conformal methods is one of marginal (or long-run) coverage. However for the purpose of Value-at-Risk Forecasting one desires conditional coverage: the probability an observation falls below the VaR level should be at most α for each time period, given the information of the past.

However it is a well known fact that conditional coverage without any distributional assumptions is unattainable in conformal prediction Lei and Wasserman (2014). Instead Vonk (2025), shows that through the integration of prior distributional information conditional coverage is achievable in conformal prediction. Hence for the estimation of the Conditional VaR, one needs to incorporate a structural prior model if one desires to achieve conditional coverage at all, and is achieved by using the probability integral transform (PIT) from the parametric model as our conformity score. This transforms the problem from finding a quantile of a time-varying distribution to finding a quantile of a (more) stable distribution in the PIT space. We extend this result to the time series setting

Proposition 3 (Finite-sample *conditional* coverage under correct full-distributional prior) *Let $\{Y_t\}_{t=1}^{T+1}$ be a time series adapted to the natural filtration $\mathscr{F}_{t-1} = \sigma(Y_1, ..., Y_{t-1})$. Suppose we have a correct prior for the conditional distribution with respect to $\mathscr{F}_{t-1}$ of Y_t through, $F_t(y) = \mathbb{P}(Y_t \leq y | \mathscr{F}_{t-1})$, and define*

$$S_t = F_t(Y_t), \quad t = 1, ..., T+1.$$

Then if split conformal calibration is applied to the transformed values S_i, by taking the order statistic $\widehat{Q} = S_{(\lceil (1-\alpha)(T+1) \rceil)}$, the resulting upper prediction band

$$C(\mathscr{F}_t) = \{y : F_{t+1}(y) \leq \widehat{Q}\}$$

satisfies

$$\mathbb{P}(Y_{T+1} \in C(\mathscr{F}_T) \mid \mathscr{F}_T) \geq 1 - \alpha.$$

We also derive an even more general result with an incorrect prior

Proposition 4 (Lower-bound Approximate Distributional Beliefs) *Let $\{Y_t\}_{t=1}^{T+1}$ be a time series adapted to the natural filtration $\mathscr{F}_{t-1} = \sigma(Y_1, \ldots, Y_{t-1})$. Fix a miscoverage level $\alpha \in (0, 1)$ and let*

$$F_t(y) = \mathbb{P}(Y_t \leq y \mid \mathscr{F}_{t-1})$$

be the true conditional CDF. Suppose we have an approximate conditional CDF $\tilde{F}_t(y)$, also adapted to $\mathscr{F}_{t-1}$ satisfying

$$\sup_{1\le t\le T+1}\sup_{y\in\mathbb{R}}\Big|\widetilde{F}_t(y)-F_t(y)\Big|\le\varepsilon \quad \textit{a.s.}$$

Define the non-conformity scores

$$\widetilde{S}_i=\widetilde{F}_t(Y_t), \quad t=1,\ldots,T.$$

and let

$$m=\lceil(1-\alpha)(T+1)\rceil, \qquad \widehat{Q}=\widetilde{S}_{(m)},$$

the m-th order statistic of the $\widetilde{S}_i$ over the calibration sample $t=1,\ldots,T$. Then the one-sided conformal prediction set

$$C(\mathscr{F}_{t-1})=\{\,y\in\mathbb{R}:\widetilde{F}_t(y)\le\widehat{Q}\}$$

satisfies, the finite-sample ***conditional*** *coverage bound*

$$\mathbb{P}\big\{Y_{T+1}\in C(\mathscr{F}_T)\ \big|\ \mathscr{F}_T\big\}\ge 1-\alpha-2\varepsilon.$$

In case the prior is perfectly specified such that the conditional distribution is iid again, and using the work of Angelopoulos, Barber, and Bates (2024), which shows that adaptive conformal with decaying step size under stationarity (i.i.d.) conditions achieves asymptotically marignal coverage, this result extends to asymptotic conditional coverage.

Note that this approach has many similarities, to the approach outlined in this paper for the Value-at-Risk estimation. Adaptive Conformal Inference tracks an empirical sample quantile of past scores. Our methodology, in contrast, directly tracks the true population quantile in the PIT-space through a consistent scoring rule. An important observation however needs to be made. While the logic of adaptive conformal Inference stems from traditional conformal methods, there is in essence nothing conformal about this approach. This observation is similarly made in Szabadváry (2024). Yet this subtle distinction in our methodology has several profound advantages.

1. Our method does not suffer from 'quantile crowding' that can occur in sample-based methods under distribution shifts, where many past scores cluster together, making the quantile tracking rigid to adapt. By targetting the α quantile directly through the quantile function, this effect is avoided.
2. Standard conformal prediction requires a conservative finite sample correction to the quantile (using the $\lceil(1-\alpha)(n+1)\rceil/n$ quantile, which is $1/(n+1)$ larger) to achieve valid coverage. Our approach has no need for such corrections, directly targetting the desired level α.
3. Due to the finite sample correction, conformal prediction struggles to produce extreme quantile estimates. as it requires at least $n\ge 1/\alpha-1$ samples to target a nominal α quantile.
4. Our framework is also able to directly estimate of the Expected shortfall at each step. This contrasts with adaptive conformal methods that can only produce conservative VaR estimates, from which the ES must be inferred through for example quantile averaging.

Potentially one advantage of Adaptive conformal methods is that in small samples when the distribution is relatively stable the quantile will converge faster (albeit conservative) to the target quantile.

Finally, while adaptive conformal methods have the appeal of being perceived as distribution-free (/assumption free). We would like to point out that given its equivalence with the proposed framework this property actually carries over, even while this is seemingly not the case as a structural prior is assumed. This can however be reconsiled realising that any map to the bounded domain $[0,1]$ is sufficient, however uninformative (e.g. always use standard normal $\mathcal{N}(0,1)$).

6 Experiments

6.1 Simulation study

6.2 Real world Data: International Equity Indices

To evaluate our proposed framework under real-world conditions, we test it by calibrating an ARMA-GARCH(1,1)-t model as the parametric prior. This prior is then wrapped with our online learning safety layers (SGD and COCOB). We conduct our analysis on the same four major international equity indices used in Patton et al. (2019): Standard & Poor's 500 (S&P500). Dow Jones Industrial Average (DJIA30), Nikkei 225 (NIK225), and the Financial Times Stock Exchange 100 Index (FTSE100). The daily log-return data spans from January 1990 to December 2016. The first 10 years of the data are used for initial estimation of the base GARCH-t model, with the subsequent 17 years reserved for out-of-sample forecasting and evaluation. All results are reported for the tail probability level of $\alpha = 0.05$.

Table 1: Summary statistics.

	S&P500	DJIA30	NIK225	FTSE100
Mean (Annualised)	6.776	7.312	-2.699	3.996
Std dev (Annualised)	17.879	17.044	24.491	17.726
Skewness	-0.244	-0.163	-0.130	-0.124
Kurtosis	8.680	8.117	5.240	5.888
VaR-0.01	-3.108	-2.992	-4.100	-3.101
VaR-0.025	-2.323	-2.182	-3.145	-2.351
VaR-0.05	-1.731	-1.639	-2.444	-1.709
VaR-0.10	-1.183	-1.126	-1.775	-1.192
ES-0.01	-4.528	-4.280	-5.699	-4.230
ES-0.025	-3.405	-3.215	-4.409	-3.294
ES-0.05	-2.697	-2.553	-3.582	-2.645
ES-0.10	-2.065	-1.955	-2.836	-2.031

The bottom panel is perhaps most insightful, revealing the workings of the adaptive safety layer. It shows the evolution of the VaR and ES parameters $(v_{u,t}, e_{u,t})$ in the bounded domain. If the GARCH-t were perfectly specified, these parameters would be constant over time. The visible drift in these parameters are direct evidence of the learning algorithm correcting for systematic errors and the misspecification of the base GARCH model in real-time. Particularly the shift around 2008 to the far end of the tail show the underestimation of

Table 2: Out-of-sample average losses and GoF p-values ($\alpha = 0.05$).

	Avg loss				VaR GoF (DQ)				ES GoF (DES)			
Model	S&P500	DJIA30	NIK225	FTSE100	S&P500	DJIA30	NIK225	FTSE100	S&P500	DJIA30	NIK225	FTSE100
RW-125	0.908	0.858	1.271	0.953	**0.058**	**0.048**	0.001	0.000	**0.116**	**0.098**	0.030	0.000
RW-250	0.959	0.909	1.286	1.002	0.003	0.002	0.024	0.000	**0.062**	0.020	0.035	0.002
RW-500	1.024	0.976	1.309	1.056	0.002	0.003	0.002	0.000	0.024	0.021	0.002	0.000
GARCH-N	0.876	0.808	1.166	0.873	0.043	**0.172**	**0.571**	0.001	0.001	0.010	**0.220**	0.000
GARCH-Skt	0.864	0.808	1.187	0.846	0.022	0.001	0.000	**0.432**	0.003	0.000	0.000	**0.350**
GARCH-EDF	0.862	0.796	1.161	0.872	0.006	0.040	**0.709**	0.000	0.023	**0.109**	**0.686**	0.000
GARCH-FZ	0.863	0.797	1.164	0.870	0.008	0.011	**0.294**	0.000	0.024	0.040	**0.377**	0.000
GAS-1F	0.853	0.784	1.171	0.889	0.007	0.002	**0.186**	0.000	**0.083**	0.021	**0.559**	0.000
SGD GARCH-t	**0.793**	**0.727**	**1.114**	**0.790**	**0.732**	**0.559**	**0.985**	**0.305**	**0.156**	**0.070**	**0.076**	0.001
COCOB GARCH-t	0.835	0.773	1.149	0.829	**0.163**	0.033	**0.510**	0.044	**0.058**	0.039	**0.484**	**0.096**

6.2.1 Analysis of Forecasting accuracy and Specification

As shown in Table in Table 2, the SGD GARCH-t model consistently achieves the lowest FZ0 loss across all four indices, indicating superior joint accuracy for VaR and ES. The COCOB GARCH-t model performs nearly as well, outperforming all other benchmarks. This demonstrates that the online safety layer effectively corrects the systematic errors of the GARCH-t prior resulting in more accurate forecasts.

The goodness of fit test confirm our motivating hypothesis that even sophisticated parametric models are inevitably misspecified. We observe that the uncorrected GARCH-t prior is rejected for every test. The GARCH-FZ and GAS-1F models show some improvement, suggesting that if one suspects the base model to be misspecified, directly estimating the model on the FZ-loss seems to be effective. In contrast to all base models, our SGD GARCH-t and COCOB GARCH-t models pass the majority of

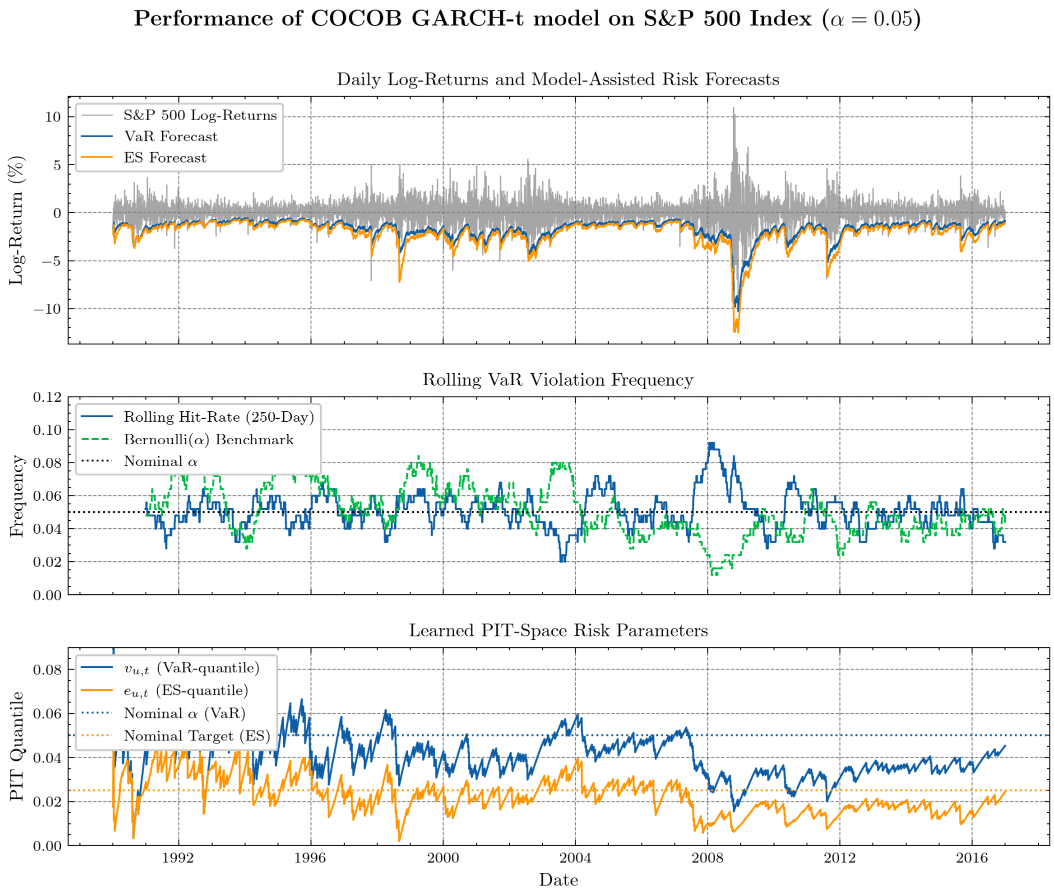

Figure 1: Performance of the COCOB GARCH-t model on the S&p 500 index ($\alpha = 0.05$). S&P500 Overview. Top Panel: Daily log-returns of the SP500 index for the out-of-sample period, overlaid with the 5% VaR and ES forecasts from the COCOB GARCH-t model. Middle Panel: The 250-day rolling average of VaR violations (hit-rate) for the COCOB GARCH-t model, compared to the nominal 5% target. Bottom Panel: The time-series of the learned 5% VaR quantile ($v_{u,t}$) and ES risk measure ($e_{u,t}$) in the bounded space.

dynamic specification tests, and the SGD GARCH-t model is not rejected by the DQ test for any of the four indices. This provides strong evidence that our main framework produces models that are more correctly specified conditionally.

6.2.2 Analysis of Unconditional and Conditional Coverage

Table 3 further reinforces these findings. The benchmark GARCH-based models (GARCH-t, GARCH-FZ, GAS-1F) exhibit empirical hit rates that are systematically too high for all four indices(e.g., 0.064 for S&P500), indicating that underestimate risk, leading to more frequent VaR violations than the nominal 5% level. This underestimation is a critical failure for risk management. Consequently, they are strongly rejected by both the Kupiec and Christoffersen tests, failing on both unconditional coverage and independence of violations. Conversely, the unconditional validity guarantees effectively kick in for our proposed models as the delivered hit rates are remarkably close to the 0.05 target. Furhtermore the Christofferson test for both our models suggest that the learning rates were well tuned for the SGD GARCH-t model, as both too large step sizes as well as too small ones cause independence of VaR violations to become dependent. We see the same thing for the COCOB GARCH-t algorithm and is particularly reassuring that the step sizes are correctly learned in a parameter free fashion.

Table 3: Out-of-sample hit rates and LR test p-values ($\alpha = 0.05$). Values in parentheses are the Universal Inference (finite-sample) equivalents.

	Hit rate				Kupiec LR p-value (UI p-value)				Christoffersen LR p-value (UI p-value)			
Model	S&P500	DJIA30	NIK225	FTSE100	S&P500	DJIA30	NIK225	FTSE100	S&P500	DJIA30	NIK225	FTSE100
RW-125	0.054	0.052	0.053	0.055	**0.206** (**0.619**)	**0.519** (**1.000**)	**0.313** (**0.725**)	**0.161** (**0.527**)	0.017 (**0.550**)	0.025 (**0.198**)	**0.203** (**1.000**)	0.000 (0.010)
RW-250	0.052	0.055	0.053	**0.051**	**0.475** (**1.000**)	**0.142** (**0.606**)	**0.313** (**1.000**)	**0.712** (**1.000**)	0.000 (**0.144**)	0.001 (**0.072**)	0.045 (**1.000**)	0.000 (**0.081**)
RW-500	0.052	0.054	0.054	0.052	**0.475** (**0.911**)	**0.260** (**0.682**)	**0.251** (**0.721**)	**0.563** (**0.911**)	0.000 (**0.252**)	0.001 (**0.499**)	**0.078** (**1.000**)	0.000 (**0.571**)
GARCH-Skt	0.040	0.036	0.032	0.053	0.003 (**0.059**)	0.000 (0.002)	0.000 (0.000)	**0.393** (**1.000**)	0.012 (**1.000**)	0.000 (**1.000**)	0.000 (**1.000**)	**0.099** (**1.000**)
GARCH-EDF	0.065	0.062	0.053	0.069	0.000 (0.012)	0.001 (0.044)	**0.313** (**0.804**)	0.000 (0.002)	0.000 (**1.000**)	0.002 (**1.000**)	**0.500** (**0.926**)	0.000 (**0.395**)
GARCH-FZ	0.064	0.064	0.056	0.070	0.000 (0.010)	0.000 (0.008)	**0.089** (**0.704**)	0.000 (0.003)	0.000 (**1.000**)	0.000 (**1.000**)	**0.154** (**0.988**)	0.000 (**0.720**)
GAS-1F	0.064	0.061	0.053	0.074	0.000 (0.007)	0.001 (0.035)	**0.348** (**0.867**)	0.000 (0.000)	0.000 (**1.000**)	0.004 (**1.000**)	**0.181** (**0.519**)	0.000 (**1.000**)
SGD GARCH-t	0.052	0.052	**0.052**	0.052	**0.613** (**1.000**)	**0.613** (**1.000**)	**0.604** (**1.000**)	**0.564** (**0.911**)	**0.865** (**1.000**)	**0.649** (**1.000**)	**0.871** (**1.000**)	**0.089** (**0.592**)
COCOB GARCH-t	**0.050**	**0.051**	**0.052**	0.052	**0.930** (**1.000**)	**0.875** (**1.000**)	**0.556** (**0.891**)	**0.661** (**1.000**)	**0.929** (**1.000**)	**0.801** (**1.000**)	**0.835** (**1.000**)	**0.141** (**0.580**)

6.2.3 Analysis of Forecast Comparison

While the average FZ0 loss and the goodness of fit tests provide evidence to suggest that our frameworks well in practise, a formal comparison is required to assess the statistical significance of the performance differences. To this end we conducted the pairwise forecast comparison for the S&P500 index. with the results presented in the Appendix 4. This table contains the 95% confidence intervals for the score difference (Row Model - Column Model), meaning that a positive difference suggests that the column model outperforms the row model. We provide both the Diebold Mariano asymptotic interval and teh finite sample, any time valid confidence sequence of Choe and Ramdas (2024).

We observe that the the models form the Model-Assisted Online Learning framework consistently and significantly outperform the vast majority of benchmarks. For instance, when comparing comparing our models against the semiparametric GARCH-FZ and one-factor GAS model of Patton et al. (2019) the resulting confidence intervals are strictly above the null of equal forecast accuracy. This provides significant evidence that our framework offers a genuine improvement. We see similar patterns comparing against the base GARCH models. Except for the GARCH-skew-t model, potentially suggesting that the fat tails incorporated in the base model already provides an improvement in forecasting ability our models cannot exploit.

We also remark that the performance of the COCOB GARCH-t model underperforms the SGD GARCH-t model. Potentially suggesting that the initial guess for the learning rate was effectively made. as we furthermore see that the SGD GARCH models dominates the other models, except the rolling window models.

Finally we remark that the decisions inferred from the confidence sequences are often in agreement with the the asymptotic confidence intervals of the DM test. Suggesting that there the assumption free framework does not lose too much power. There are however also some cases where the DM test rejects equal forecasting accuracy while the anytime-valid test is in disagreement.

6.2.4 Economic Analysis and Model Diagnostics of the 2008 Financial Crisis

Beyond aggregate performance metrics, the strength of our "reality-centric" framework is its ability to serve as a live diagnostic tool on the health of the underlying base model. The evolution of the learned VaR and ES parameters in the bounded PIT-space, $(v_{u,t}, e_{u,t})$, allows us to see how well the structural prior (the GARCH-t model) is capturing the changing market dynamics. If the prior were perfectly specified, these parameters would remain constant over time. Any persistent drift, therefore, is an immediate signal of model misspecification, allowing for a real-time economic analysis of the model's failings.

To illustrate this capability, we zoom in on the period surrounding the 2008 Global Financial Crisis, a time of high market volatility and structural change in market dynamics. Figure 6.2.4 presents the daily log-returns of the S&P 500 and the learned parameters from our COCOB GARCH-t model for the period 2006-2010 in the bounded domain.

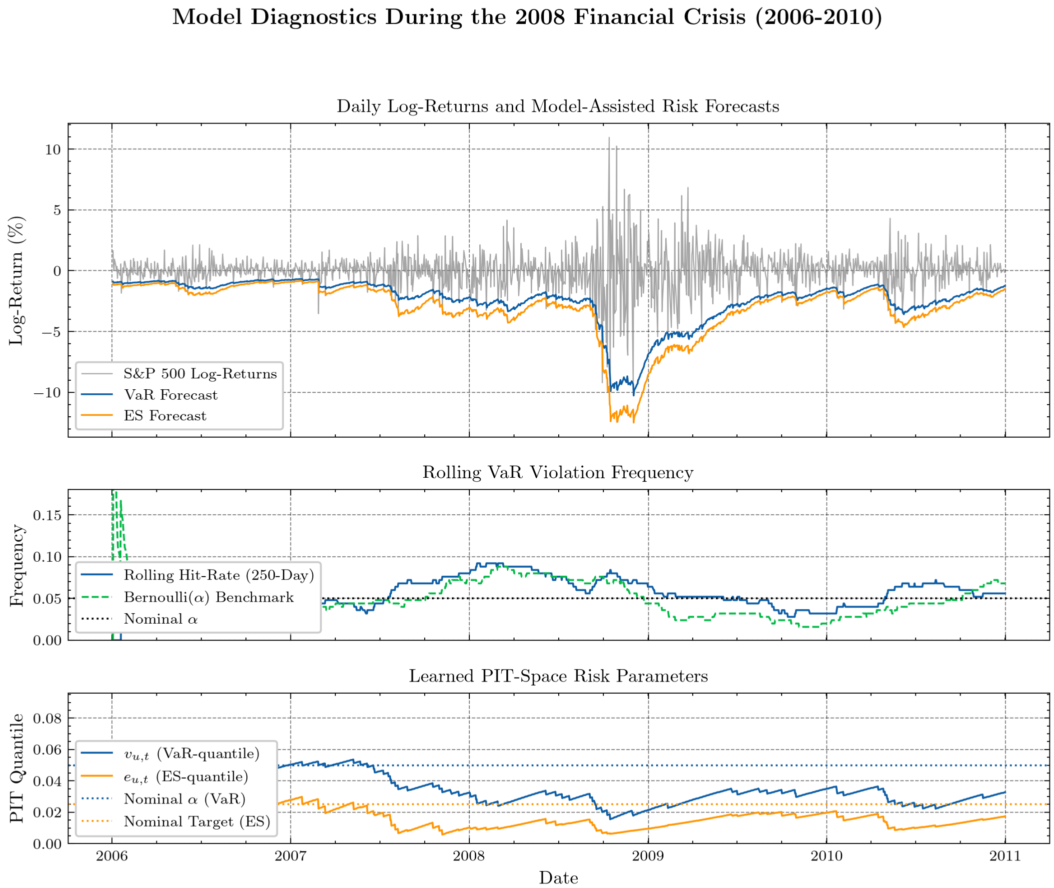

Figure 2: Model Diagnostics During the 2008 Financial Crisis (2006-2010). Bottom Panel: The time-series of the learned 5% VaR quantile ($v_{u,t}$) and ES risk measure ($e_{u,t}$) in the bounded domain from the COCOB GARCH-t model.

The analysis of this period shows a few things. In the relatively calm period before the financial crisis, up until the start of 2007 we observe that the learned parameters $(v_{u,t}, e_{u,t})$ are largely stable and close to the nominal targets. were the model correctly specified. this indicates that the GARCH-t prior was a reasonably adequate approximation of market dynamics. However, as the crisis began to unfold in late 2007 and later intensified throughout 2008, we observe a downward drift in both parameters and remains there for two years for the ES, but stays more persistently low for the VaR. This drift shows the adaptive safety layer working in real time. It signals that the base GARCH-t model was systematically underestimating the true tail risk during the crisis. Furthermore the widening of the parameters in 2007-2008 indicates increased fat tails. Later this gap converges again, suggesting that the model might not only underestimate the overall volatility rather than the fat-tailed nature of the returns distribution that characterized this period.

The economic interpretation is direct as it shows that the COCOB algorithm was continuously indicating that the base model's forecasts were not conservative enough, and it was consistently being proven right. The safety layer was therefore forced to aggressively push the effective quantiles further into the tail to correct for the base model's failure to adapt. This analysis shows the benefit of our reality-centric framework not only does it provide better forecasts during a crisis but also delivers an interpretable, real-time diagnostic that shows precisely how a standard theory based model is failing.

Such analysis can aid financial risk managers understand the nature of the model's misspecification as it happens. This capability of providing both safety and insight is once more a quality that adds to the transparency and trustworthiness of financial risk management systems.

7 Discussion & Conclusion

At its core, this paper introduces and advocates for a novel "reality-centric" framework for reliable forecasting. This philosophy proposes that predictive systems in high-stakes environments, must not only be effective but also demonstrably reliable and accountable, especially given the inherent and unavoidable misspecification of models. Our approach, Model-Assisted Online Learning, adheres to this philosophy by first leveraging established domain knowledge through a base model, and then applying an adaptive online learning layer to correct for its inevitable misspecifications in real time.

Experiment conducted on international equity indices provide strong support for our framework and its extension of the research by Patton et al. (2019). Across all tested datasets, our proposed SGD GARCH-t and COCOB GARCH-t models consistently outperformed benchmark models, including the semiparametric approach of Patton et al. (2019). Specifically, our models achieved superior forecasting accuracy. Furthermore, the theoretical guarantees of unconditional coverage seem to be effective, and allow the models to pass specification tests, as evidenced by the goodness-of-fit tests, and highlight the framework's capacity to adapt to distributional shifts. Crucially, while similarly estimating the base models directly on the FZ loss, our work extends Patton, Ziegel, and Chen (2019) by providing robust guarantees even when dynamics are misspecified, which allows for the advancement for real-world applicability. Furthermore, we show that our framework has strong connections to adaptive conformal inference, but our approach offers distinct advantages, such as avoiding "quantile crowding," eliminating the need for conservative finite sample corrections, and allowing direct estimation of Expected Shortfall.

However, certain considerations regarding our methodology warrant further discussion. A key aspect of our approach involves transforming the returns to a bounded space, which we to chose perform using the cumulative density function. We noted that directly estimating the VaR and ES on this bounded sequence, may not yield the *true* expected shortfall due to the curvature of the CDF (Jensen's inequality). This was a deliberate approximation for simplicity, however we would like to bring to the attention that even if the transformed sequence is IID uniform (in the most optimal case for the CDF mapping), the true ES in the bounded domain does not necessarily remain stable due to the possibly changing curvature of the conditional CDF over time. Therefore, in order for a stable ES in the case of a correct specification, or any functional for that matter, one needs to use alternative transformations, based on the conditional CDF, that are explicitly designed to preserve such stability. This is an important area for further research.

It is also important to acknowledge the nature of the safety guarantees provided by this framework. Our method delivers unconditional, long-run validity. This means that over a sufficiently long period, the forecasts have the same properties the correct risk measures would have, regardless of how the underlying data or base model behaves. We argue that this is a powerful and a valuable guarantee to satisfy regulatory requirements or sustain performance in unpredictable real-world scenarios. However, we also recognise that future research in reality centric methods could strive for even stricter notions of safety. For instance, finite-sample or strong conditional guarantees, which ensure validity at every single time step. This would be a highly desirable guarantee that is strongly sought after but currently only comes with stronger assumptions about the data generating process.

Crucially, this paper introduces the 'reality-centric' framework as a novel approach to reliable forecasting. We would like to emphasise that the fields of online learning theory and game theory seem to be interesting areas of research for tackling the complexities of the real world, as they are inherently designed for scenarios with minimal prior assumptions or even adversarial conditions. We have observed that many concepts between these areas seem to fulfil many purposes. For example, adaptations of Kelly betting were employed in the COCOB algorithm for optimal step sizes but also in the sequential forecast comparison framework of Choe and Ramdas (2024) for optimal evidence aggregation. By drawing upon these robust, adaptive, and assumption-light principles, we aim to establish a new direction for developing trustworthy predictive systems in complex, real-world scenarios.

In conclusion, the Model-Assisted Online Learning framework represents a significant step towards truly reality-centric forecasting. By integrating domain expertise with the robust, adaptive, and assumption-light principles of online learning and game theory, we offer a method that is not only effective but also demonstrably reliable and accountable, addressing critical limitations of existing approaches in financial risk management.

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A Proofs

For any sequence of observations $\{u_t\}_{t=1}^T$ and for any constant step size $\eta > 0$, the sequence of VaR estimates $\{q_t\}$ produced by the VaR update rule satisfies

$$\left| \frac{1}{T} \sum_{t=1}^{T} (\mathbb{1}\{u_t \leq v_{u,t}\} - \alpha) \right| \leq \frac{D}{\eta T}$$

Therefore, in particular

$$\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \mathbb{1}\{u_t \leq v_{u,t}\} \overset{\text{a.s}}{=} \alpha$$

Proof: Start with the update rule

$$v_{t+1} = v_t + \eta(\alpha - \text{err}_t).$$

Rearanging

$$\text{err}_t - \alpha = -\frac{v_{t+1} - v_t}{\eta}$$

summing from $t = 1$ to T

$$\sum_{t=1}^{T} (\text{err}_t - \alpha) = \sum_{t=1}^{T} \left(-\frac{v_{t+1} - v_t}{\eta} \right)$$

notice that $\sum_{t=1}^{T} (v_{t+1} - v_t) = (v_2 - v_1) + (v_3 - v_2) + \cdots + (v_{T+1} - v_T) = v_{T+1} - v_1$. Thus we are left with

$$\sum_{t=1}^{T} (\text{err}_t - \alpha) = \frac{v_1 - v_{T+1}}{\eta}$$

Now taking the absolute value and diving by T on both sides

$$\left| \frac{1}{T} \sum_{t=1}^{T} (\text{err}_t - \alpha) \right| = \frac{|v_1 - v_{T+1}|}{\eta T}$$

By assumption of our bounded domain, or simply the projection operator we know that $|v_1 - v_{T+1}| \leq D$. substituing this back we arrive at the final result

$$\left| \frac{1}{T} \sum_{t=1}^{T} (\mathbb{1}\{u_t \leq v_{u,t}\} - \alpha) \right| \leq \frac{D}{\eta T}$$

$\square$

Proposition 5 (ES Weak Unconditional Validity) *For any sequence of observations $\{u_t\}_{t=1}^T$ and for any constant step size $\eta_e > 0$, the sequence of ES estimates produced by the US update rule satisfies*

$$\left| \frac{1}{T} \sum_{t=1}^{T} \left[e_{u,t} - v_{u,t} + \frac{v_{u,t} - u_t}{\alpha} \mathbb{1}\{u_t \leq q_{u,t}\} \right] \right| \leq \frac{D}{\eta_e T}$$

Therefore, as

$$\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} e_{u,t} \overset{a.s}{=} \lim_{t\to\infty} \frac{1}{T} \sum_{t=1}^{T} u_t \mathbb{1}\{u_t \leq v_{u,t}\}$$

Proof: The start it the same as proof A

This gives that

$$\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \left(e_{u,t} - v_{u,t} + \frac{v_{u,t} - u_t}{\alpha} \mathbb{I}\{u_t \leq v_{u,t}\} \right) = 0$$

We can rearrange this expression to solve for the long-run average of $e_{u,t}$:

$$\lim_{T\to\infty}\frac{1}{T}\sum_{t=1}^{T}e_{u,t}=\lim_{T\to\infty}\frac{1}{T}\sum_{t=1}^{T}\left(v_{u,t}-\frac{v_{u,t}-u_t}{\alpha}\mathbb{I}\{u_t\le v_{u,t}\}\right)$$

expandin the right-hand side:

$$=\lim_{T\to\infty}\left[\frac{1}{T}\sum_{t=1}^{T}v_{u,t}\left(1-\frac{\mathbb{I}\{u_t\le v_{u,t}\}}{\alpha}\right)+\frac{1}{T\alpha}\sum_{t=1}^{T}u_t\mathbb{I}\{u_t\le v_{u,t}\}\right]$$

Now, we analyze the two terms in this sum separately. For the first term, we use proof A, which guarantees that the long-run frequency of VaR violations converges to α:

$$\lim_{T\to\infty}\frac{1}{T}\sum_{t=1}^{T}\mathbb{1}\{u_t\le v_{u,t}\}=\alpha$$

By Slutsky the term inside the parentheses, $(1-\frac{\mathbb{I}\{u_t\le v_{u,t}\}}{\alpha})$, has a long-run average of zero. The entire first term converges to zero. We are therefore left with the second term:

$$\lim_{T\to\infty}\frac{1}{T}\sum_{t=1}^{T}e_{u,t}=\lim_{T\to\infty}\frac{1}{T\alpha}\sum_{t=1}^{T}u_t\mathbb{1}\{u_t\le v_{u,t}\}$$

which completes our proof. □

B Conditional Coverage Under Correct Prior Beliefs

In this section, we formally show that if the prior belief $\hat{F}_t(Y_t)$ is the true conditional distribution, the conformal approach using the conformity score $S(Y)=F_t(y)$ achieves exact conditional coverage.

Proposition 6 (Finite-sample *conditional* coverage under correct full-distributional prior) *Let $\{Y_t\}_{t=1}^{T+1}$ be a time series adapted to the natural filtration $\mathscr{F}_{t-1}=\sigma(Y_1,...,Y_{t-1})$. Suppose we have a correct prior for the conditional distribution with respect to $\mathscr{F}_{t-1}$ of Y_t through, $F_t(y)=\mathbb{P}(Y_t\le y|\mathscr{F}_{t-1})$, and define*

$$S_t=F_t(Y_t),\quad t=1,...,T+1.$$

Then if split conformal calibration is applied to the transformed values S_i, by taking the order statistic $\widehat{Q}=S_{(\lceil(1-\alpha)(T+1)\rceil)}$, the resulting upper prediction band

$$C(\mathscr{F}_t)=\{y:F_{t+1}(y)\le\widehat{Q}\}$$

satisfies

$$\mathbb{P}(Y_{T+1}\in C(\mathscr{F}_T)\mid\mathscr{F}_T)\ge 1-\alpha.$$

Proof: By the probability-integral transform and correctness of the prior, each $S_t=F_t(Y_t)\sim\text{Unif}(0,1)$. Furthermore, as the prior is correctly specified a priori, the $\{S_1,\dots,S_{T+1}\}$ are i.i.d. Uniform$(0,1)$. Then exchangeability implies the standard split conformal result

$$\mathbb{P}(S_{T+1}\le\widehat{Q})\ge 1-\alpha.$$

But since $\widehat{Q}$ is a function of $S_1,...,S_T$ which are independent of S_{T+1}, this inequality holds conditionally on $\mathscr{F}_T$: $\mathbb{P}(S_{T+1}\le\widehat{Q}\mid\mathscr{F}_T)=\mathbb{P}(S_{T+1}\le\widehat{Q})$, so in fact $\mathbb{P}(S_{T+1}\le\widehat{Q}\mid\mathscr{F}_T)\ge 1-\alpha$. Finally, setting $\widehat{Q}_Y(\mathscr{F}_{t-1})=F_t^{-1}(\widehat{Q})\}$, so that that $\{Y_{T+1}\le\widehat{Q}_Y(\mathscr{F}_T)\}=\{F_{T+1}(Y_{T+1})\le\widehat{Q}\}$, maps the coverage bound back to the Y-space. Concluding that,

$$\mathbb{P}(Y_{T+1}\le C(\mathscr{F}_T)\mid\mathscr{F}_T)\ge 1-\alpha.$$

□

Lemma 1 *Let $S_1,\dots S_T$ and $\widetilde{S}_1,\dots,\widetilde{S}_T$ be real numbers such that*

$$|S_t-\widetilde{S}_t|\le\varepsilon\quad\textit{for all } t=1,\dots,T.$$

And $S_{(1)}\le...\le S_{(T)}$ and $\widetilde{S}_{(1)}\le...\le\widetilde{S}_{(T)}$ be the order statistics of $S_1,\dots S_T$ and $\widetilde{S}_1,\dots,\widetilde{S}_T$ respectively. Then for every $m=1,...,T$,

$$|S_{(m)}-\widetilde{S}_{(m)}|\le\varepsilon$$

Proof: There are m, S_t such that $S_t \leq S_{(m)}$, and for each t we have

$$\widetilde{S}_t \leq S_t + \varepsilon \leq S_{(m)} + \varepsilon$$

which yields

$$\widetilde{S}_{(m)} \leq S_{(m)} + \varepsilon.$$

Doing the same for the lower-bound proves the statement. □

Proposition 7 (Lower-bound Approximate Distributional Beliefs) *Let $\{Y_t\}_{t=1}^{T+1}$ be a time series adapted to the natural filtration $\mathscr{F}_{t-1} = \sigma(Y_1, \dots, Y_{t-1})$. Fix a miscoverage level $\alpha \in (0,1)$ and let*

$$F_t(y) = \mathbb{P}(Y_t \leq y \mid \mathscr{F}_{t-1})$$

be the true conditional CDF. Suppose we have an approximate conditional CDF $\tilde{F}_t(y)$, also adapted to $\mathscr{F}_{t-1}$ satisfying

$$\sup_{1 \leq t \leq T+1} \sup_{y \in \mathbb{R}} \left| \widetilde{F}_t(y) - F_t(y) \right| \leq \varepsilon \quad \textit{a.s.}$$

Define the non-conformity scores

$$\widetilde{S}_i = \widetilde{F}_t(Y_t), \quad t = 1, \dots, T.$$

and let

$$m = \lceil (1-\alpha)(T+1) \rceil, \qquad \widehat{Q} = \widetilde{S}_{(m)},$$

the m-th order statistic of the $\widetilde{S}_i$ over the calibration sample $t = 1, \dots, T$. Then the one-sided conformal prediction set

$$C(\mathscr{F}_{t-1}) = \{\, y \in \mathbb{R} : \widetilde{F}_t(y) \leq \widehat{Q} \}$$

*satisfies, the finite-sample **conditional** coverage bound*

$$\mathbb{P}\big\{ Y_{T+1} \in C(\mathscr{F}_T) \;\big|\; \mathscr{F}_T \big\} \geq 1 - \alpha - 2\varepsilon.$$

Proof:

(1) By assumption, for every t and every realization $Y_1, \dots, Y_t$, the true scores $S_i = F_t(Y_t)$ and the approximate scores $\widetilde{S}_t = \widetilde{F}_t(Y_t)$ satisfy:

$$\left| \widetilde{S}_i - S_i \right| = \left| \widetilde{F}_t(Y_t) - F_t(Y_t) \right| \leq \varepsilon.$$

Thus, all "tilde" scores lie within ε of their ideal uniform counterparts.

(2) If conformal calibration were applied to the true scores $S_1, \dots, S_T$, which are i.i.d. Uniform(0,1), then setting

$$Q^* = S_{(m)}, \quad \text{where } m = \lceil (1-\alpha)(T+1) \rceil,$$

Split Conformal Prediction implies the marginal guarantee

$$\mathbb{P}(S_{T+1} \leq Q^*) \geq \frac{m}{T+1} \geq 1 - \alpha.$$

Furthermore, From theorem 1, because the S_t are i.i.d. Uniform(0,1) and thus their distribution is independent of $\mathscr{F}_{t-1}$, this guarantee holds conditionally as well:

$$\mathbb{P}(S_{T+1} \leq Q^* \mid \mathscr{F}_T) = \mathbb{P}(S_{T+1} \leq Q^*) \geq 1 - \alpha.$$

(3) Since each $\widetilde{S}_t$ is within ε of S_t, their m-th order statistics satisfy:

$$Q^* - \varepsilon \leq \widetilde{S}_{(m)} = \widehat{Q} \leq Q^* + \varepsilon.$$

This follows because shifting all T points in a dataset by at most ε can move any order-statistic by at most ε. See lemma 1.

(4) We want to lower-bound the conditional coverage probability:

$$\begin{aligned}
\mathbb{P}\big(Y_{T+1} \in C(\mathscr{F}_T) \mid \mathscr{F}_T\big) &= \mathbb{P}\big(\widetilde{F}_t(Y_{n+1}) \leq \widetilde{Q} \mid \mathscr{F}_T\big) = \mathbb{P}\big(\widetilde{S}_{T+1} \leq \widehat{Q} \mid \mathscr{F}_T\big) \\
&\geq \mathbb{P}\big(S_{T+1} \leq \widehat{Q} - \varepsilon \mid \mathscr{F}_T\big) \quad \text{(since } \widetilde{S}_{n+1} \leq S_{n+1} + \varepsilon) \\
&\geq \mathbb{P}\big(S_{T+1} \leq (Q^* - \varepsilon) - \varepsilon \mid \mathscr{F}_T\big) \quad \text{(since } \widehat{Q} \geq Q^* - \varepsilon) \\
&= \mathbb{P}\big(S_{T+1} \leq Q^* - 2\varepsilon \mid \mathscr{F}_T\big)
\end{aligned}$$

But $S_{T+1} \sim \text{Uniform}(0, 1)$ given $\mathscr{F}_T$. Hence

$$\begin{aligned}
\mathbb{P}\big(S_{T+1} \leq Q^* - 2\varepsilon \mid \mathscr{F}_T\big) &= \mathbb{E}_{Q^*}\big[\mathbb{P}\big(S_{T+1} \leq Q^* - 2\varepsilon \mid \mathscr{F}_T, Q^*\big)\big] \\
&= \mathbb{E}_{Q^*}\big[(Q^* - 2\varepsilon)_+\big] \geq \mathbb{E}_{Q^*}\big[Q^* - 2\varepsilon\big] \quad \text{(Jensen's inequality)} \\
&= \mathbb{E}_{Q^*}\big[Q^*\big] - 2\varepsilon = \frac{m}{T+1} - 2\varepsilon \geq 1 - \alpha - 2\varepsilon.
\end{aligned}$$

□

C Results Forecast Comparison

Table 4: Pairwise Forecast Comparison using the FZ0 Score (S&P 500, $\alpha = 0.05$)

Row Model	**Column Model**										
	COCOB	**G-FZ**	**GAS-1F**	**G-EDF**	**G-N**	**G-Skt**	**Hybrid**	**RW-125**	**RW-250**	**RW-500**	**SGD**
COCOB	—	CS: [-0.30, -0.17]* CI: [-0.26, -0.20]*	CS: [-0.39, -0.19]* CI: [-0.34, -0.25]*	CS: [-0.28, -0.16]* CI: [-0.25, -0.20]*	CS: [-0.27, -0.15]* CI: [-0.24, -0.19]*	CS: [-0.04, 0.05] CI: [-0.02, 0.02]	CS: [-0.32, -0.16]* CI: [-0.27, -0.20]*	CS: [-0.22, 0.08] CI: [-0.14, -0.00]*	CS: [-0.25, 0.13] CI: [-0.15, 0.03]	CS: [-0.30, 0.21] CI: [-0.15, 0.06]	CS: [0.01, 0.12]* CI: [0.04, 0.08]*
G-FZ	CS: [0.17, 0.30]* CI: [0.20, 0.26]*	—	CS: [-0.13, 0.01] CI: [-0.09, -0.03]*	CS: [-0.00, 0.02] CI: [0.00, 0.02]*	CS: [0.01, 0.03]* CI: [0.01, 0.03]*	CS: [0.20, 0.27]* CI: [0.22, 0.25]*	CS: [-0.05, 0.04] CI: [-0.02, 0.02]	CS: [0.00, 0.32]* CI: [0.10, 0.23]*	CS: [-0.00, 0.35] CI: [0.09, 0.25]*	CS: [-0.04, 0.42] CI: [0.09, 0.29]*	CS: [0.20, 0.40]* CI: [0.26, 0.33]*
GAS-1F	CS: [0.19, 0.39]* CI: [0.25, 0.34]*	CS: [-0.01, 0.13] CI: [0.03, 0.09]*	—	CS: [-0.00, 0.15] CI: [0.04, 0.10]*	CS: [0.00, 0.16]* CI: [0.05, 0.11]*	CS: [0.21, 0.38]* CI: [0.26, 0.33]*	CS: [0.00, 0.11]* CI: [0.04, 0.08]*	CS: [0.06, 0.39]* CI: [0.15, 0.29]*	CS: [0.05, 0.41]* CI: [0.15, 0.31]*	CS: [0.02, 0.48]* CI: [0.15, 0.34]*	CS: [0.24, 0.48]* CI: [0.31, 0.40]*
G-EDF	CS: [0.16, 0.28]* CI: [0.20, 0.25]*	CS: [-0.02, 0.00] CI: [-0.02, -0.00]*	CS: [-0.15, 0.00] CI: [-0.10, -0.04]*	—	CS: [0.01, 0.01]* CI: [0.01, 0.01]*	CS: [0.19, 0.26]* CI: [0.21, 0.24]*	CS: [-0.06, 0.03] CI: [-0.03, 0.01]	CS: [0.00, 0.30]* CI: [0.09, 0.22]*	CS: [-0.01, 0.33] CI: [0.08, 0.24]*	CS: [-0.05, 0.41] CI: [0.08, 0.28]*	CS: [0.19, 0.39]* CI: [0.25, 0.32]*
G-N	CS: [0.15, 0.27]* CI: [0.19, 0.24]*	CS: [-0.03, -0.01]* CI: [-0.03, -0.01]*	CS: [-0.16, -0.00]* CI: [-0.11, -0.05]*	CS: [-0.01, -0.01]* CI: [-0.01, -0.01]*	—	CS: [0.18, 0.25]* CI: [0.20, 0.23]*	CS: [-0.07, 0.02] CI: [-0.04, -0.00]*	CS: [-0.01, 0.29] CI: [0.08, 0.21]*	CS: [-0.02, 0.32] CI: [0.07, 0.23]*	CS: [-0.06, 0.40] CI: [0.07, 0.27]*	CS: [0.18, 0.38]* CI: [0.24, 0.31]*
G-Skt	CS: [-0.05, 0.04] CI: [-0.02, 0.02]	CS: [-0.27, -0.20]* CI: [-0.25, -0.22]*	CS: [-0.38, -0.21]* CI: [-0.33, -0.26]*	CS: [-0.26, -0.19]* CI: [-0.24, -0.21]*	CS: [-0.25, -0.18]* CI: [-0.23, -0.20]*	—	CS: [-0.30, -0.18]* CI: [-0.26, -0.21]*	CS: [-0.22, 0.08] CI: [-0.14, -0.00]*	CS: [-0.23, 0.10] CI: [-0.15, 0.02]	CS: [-0.28, 0.18] CI: [-0.15, 0.06]	CS: [-0.01, 0.13] CI: [0.04, 0.08]*
Hybrid	CS: [0.16, 0.32]* CI: [0.20, 0.27]*	CS: [-0.04, 0.05] CI: [-0.02, 0.02]	CS: [-0.11, -0.00]* CI: [-0.08, -0.04]*	CS: [-0.03, 0.06] CI: [-0.01, 0.03]	CS: [-0.02, 0.07] CI: [0.00, 0.04]*	CS: [0.18, 0.30]* CI: [0.21, 0.26]*	—	CS: [0.03, 0.31]* CI: [0.11, 0.23]*	CS: [0.01, 0.34]* CI: [0.10, 0.25]*	CS: [-0.02, 0.41] CI: [0.10, 0.29]*	CS: [0.20, 0.40]* CI: [0.26, 0.34]*
RW-125	CS: [-0.08, 0.22] CI: [0.00, 0.14]*	CS: [-0.32, -0.00]* CI: [-0.23, -0.10]*	CS: [-0.39, -0.06]* CI: [-0.29, -0.15]*	CS: [-0.30, -0.00]* CI: [-0.22, -0.09]*	CS: [-0.29, 0.01] CI: [-0.21, -0.08]*	CS: [-0.08, 0.22] CI: [0.00, 0.14]*	CS: [-0.31, -0.03]* CI: [-0.23, -0.11]*	—	CS: [-0.10, 0.11] CI: [-0.04, 0.06]	CS: [-0.14, 0.19] CI: [-0.05, 0.10]	CS: [-0.04, 0.31] CI: [0.05, 0.21]*
RW-250	CS: [-0.13, 0.25] CI: [-0.03, 0.15]	CS: [-0.35, 0.00] CI: [-0.25, -0.09]*	CS: [-0.41, -0.05]* CI: [-0.31, -0.15]*	CS: [-0.33, 0.01] CI: [-0.24, -0.08]*	CS: [-0.32, 0.02] CI: [-0.23, -0.07]*	CS: [-0.10, 0.23] CI: [-0.02, 0.15]	CS: [-0.34, -0.01]* CI: [-0.25, -0.10]*	CS: [-0.09, 0.11] CI: [-0.06, 0.04]	—	CS: [-0.08, 0.12] CI: [-0.03, 0.07]	CS: [-0.09, 0.34] CI: [0.03, 0.22]*
RW-500	CS: [-0.21, 0.30] CI: [-0.06, 0.15]	CS: [-0.42, 0.04] CI: [-0.29, -0.09]*	CS: [-0.48, -0.02]* CI: [-0.34, -0.15]*	CS: [-0.41, 0.05] CI: [-0.28, -0.08]*	CS: [-0.40, 0.06] CI: [-0.27, -0.07]*	CS: [-0.18, 0.28] CI: [-0.06, 0.15]	CS: [-0.41, 0.02] CI: [-0.29, -0.10]*	CS: [-0.19, 0.14] CI: [-0.10, 0.05]	CS: [-0.12, 0.08] CI: [-0.07, 0.03]	—	CS: [-0.16, 0.38] CI: [-0.01, 0.22]
SGD	CS: [-0.12, -0.01]* CI: [-0.08, -0.04]*	CS: [-0.40, -0.20]* CI: [-0.33, -0.26]*	CS: [-0.48, -0.24]* CI: [-0.40, -0.31]*	CS: [-0.39, -0.19]* CI: [-0.32, -0.25]*	CS: [-0.38, -0.18]* CI: [-0.31, -0.24]*	CS: [-0.13, 0.01] CI: [-0.08, -0.04]*	CS: [-0.40, -0.20]* CI: [-0.34, -0.26]*	CS: [-0.31, 0.04] CI: [-0.21, -0.05]*	CS: [-0.34, 0.09] CI: [-0.22, -0.03]*	CS: [-0.38, 0.16] CI: [-0.22, 0.01]	—

Notes: This table presents pairwise forecast comparison results based on the FZ0 score for the S&P 500 index. Higher scores indicate better performance. Each cell shows the 95% confidence interval for the score difference (Row Model - Column Model). CS is the anytime-valid Confidence Sequence from Choe & Ramdas (2024). CI is the classical Diebold-Mariano asymptotic confidence interval. An asterisk (*) indicates the interval does not contain zero, implying a statistically significant difference at the 5% level. Model abbreviations are as follows: COCOB (COCOB GARCH-t), G-FZ (GARCH-FZ), GAS-1F (GAS-1F), G-EDF (GARCH-EDF), G-N (GARCH-Normal), G-Skt (GARCH-Skewed-t), Hybrid (Hybrid), RW (Rolling Window), SGD (SGD GARCH-t).

D Bias due to estimating the ES in the bounded domain

Table 5: Bias for Standard Normal Distribution

α	True ES	Biased ES	Bias	Relative Bias (%)
5.0%	-2.063	-1.960	0.103	4.981

Table 6: Bias for Student's-t Distribution

DoF	α	True ES	Biased ES	Bias	Relative Bias (%)
3	5.0%	-3.874	-3.182	0.692	17.857
5	5.0%	-2.890	-2.571	0.320	11.057
10	5.0%	-2.408	-2.228	0.180	7.485
30	5.0%	-2.166	-2.042	0.124	5.712

E Code Reproducibility

To ensure the full reproducibility of the results presented in this thesis, all associated code, data, and analysis scripts are provided in a single zip file submitted alongside this document. The code is implemented in Python and uses mostly standard libraries, however for the confidence sequences for which code has been used from Choe and Ramdas (2024) a seperate `C++` compiler needs to be installed.

The provided zip file contains a folder including the raw time-series data for the four international equity indices used in the empirical analysis (S&P 500, DJIA30, NIK225, FTSE100) obtained from datastream. of which the S&P500 data comes directly from the github page of Patton et al. (2019).

The core implementation of the Model-Assisted Online Learning framework is organized into a Python module named `thesis_code`. This module contains separate sub-modules for each component of the framework such as the implementation of an ARMA-GARCH-t base model. A online learning module Containing the implementation of the online learning safety layers, including both the standard SGD GARCH-t and the parameter-free COCOB GARCH-t Algorithms which is adapted from the github page of Orabona and Tommasi (2017).

It also contains a simulation file containing the code for generating simulated data from an ARMA GARCH model. There is a module containing the functions for running the goodness-of-fit and forecast comparison tests, including the Split Likelihood Ratio tests and the Diebold-Mariano test. Then the `replication_code_final.ipynb` contains a notebook that replicates the results of Patton et al. (2019) based on the data originating from Datastream. Furthermore the primary script used to generate all the results, tables, and figures for our proposed models presented in this thesis requires the user to modify desired index (e.g., 'SP500', 'DJIA30', etc.) at the start to execute the script. The code will then automatically load the appropriate data, run the full analysis.